

Handheld Test And Measurement Devices Growing to The Standards of Benchtops



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The continuously-evolving technology, modification and creation of standards, and increasing requirements of customers are pushing the test and measurement equipment industry to move at a faster pace. All players in this domain, whether big or small, are striving to manufacture and supply devices that fulfil these criteria. Traditional stand-alone test equipment are usually heavy, need to be connected to the line, and often do not work well in rugged outdoors, hard-to-reach or moving environments. This has led to one notable trend in this segment—increasing demand for handheld devices, observes Naresh Narasimhan of Tektronix India.

The myth about handhelds

People often perceive handhelds to be inferior to benchtops in terms of functionality, performance, precision and accuracy.

Handheld test and measurement devices



But, if you look at today's test and measurement industry, many models available in the market offer very good performance as compared to benchtop instruments. "Now, a handheld instrument is featured with full specifications that are able to meet specific end-user application requirements," says Madhukar Tripathi of Anritsu India. "They offer better cost-performance to buyers," he adds.

When and why to use handhelds

There are three main scenarios where test and measurement is done on the field. The first is during installation, where technicians or engineers use the device to verify whether the system is up-and-running with the right frequency, voltage or such similar parameters. The second situation is during periodic, condition based or corrective maintenance of the system. This is where you need to assess the performance, recalibrate the system, identify and rectify the fault, and prevent possible issues that may come up. Third, it is used for system monitoring. An example is the monitoring of spectrum to check whether a band is available or not.

Handheld devices are mostly used in field applications because such cases require portability. The device should have battery-operation capability and the battery should provide enough backup. "High-accuracy and repeatability are desired, but not mandatory for field devices," says Ashish Jain of Keysight India, "because most field applications are a go/no-go test where the user wants to know whether the system is functioning or not."

Where the industry is heading

Listed below are some general trends we find in the handheld test and measurement industry.

Improved measurement range. Elec-